Accelerating Diffusion Sampling with Birth-Death Dynamics

An Empirical Evaluation and Diagnostic Study

IE 643: Deep Learning - Theory and Practice (Course Project)

Autumn Semester 2026

Team Name: idk **Team Roll Nos.:** 24B0928, 24B1052

IIT Bombay

code link - github.com/Aryaman-Prasad/Birth-Death-acceleration-of-diffusion-sampling

Presentation Outline

Part I: Problem & Methodology

- **Problem Statement:** Diffusion sampling is slow at inference.
- **Workflow:** Step-by-step project progression.
- **Before Prep Review:** Math setup, coordinate scaling, 2D toy ring.
- **After Prep Review:** Image scaling, log-density wall, evaluation metrics.
- **Proposed Approach:** Operator splitting, branching math, U-Net architecture.

Part II: Experiments & Takeaways

- **Datasets & Setup:** 2D Ring, CIFAR-10, CelebA-HQ faces.
- **Results I:** Synthetic Langevin mixing vs. diffusion steps.
- **Results II:** High-dimensional diagnostics & biased score correction.
- **Results III:** CIFAR-10 & CelebA-HQ benchmarks.
- **Conclusions:** What worked, what failed, and why.
- **Plan for Novelty Assessment:** 2 concrete extensions.

Task & Problem Statement

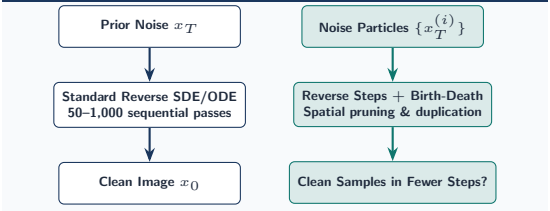
The Inference Bottleneck in Diffusion

- High visual fidelity, but sampling requires **50 to 1,000 sequential passes** through a heavy U-Net (> 35M parameters).
- Sequential noise chipping makes real-time deployment impractical.

The Core Research Question

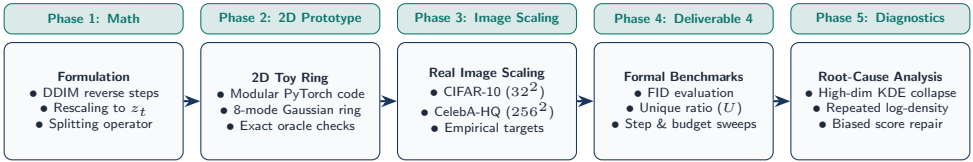
Can we add continuous-time Birth-Death dynamics to diffusion sampling to prune low-density particles, duplicate promising ones, and reach clean samples in fewer steps?

Comparing Standard Diffusion vs. Our Premise



Inspired by Lu, Lu & Nolen (2019): BD accelerates MCMC across energy barriers.

Project Workflow: Problem Statement to Completion



Execution Strategy: Theoretical derivations → 2D controlled verification → cloud GPU scaling on real images → standardized metrics → targeted diagnostics.

Work Done Before Prep Review: Math & Rescaling

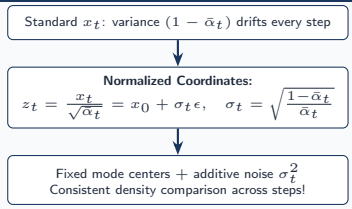
1. Reverse Diffusion Steps (DDIM)

- **Forward Process:**
  $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$.
- **Model Target:** Neural network predicts noise
  $\epsilon_\theta(x_t, t) \propto -\nabla \log p_t(x_t)$.
- **Reverse Transition:**

$$x_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}}\hat{x}_0 + c_1\epsilon_\theta(x_{\tau_i}) + \sigma_{\tau_i}\epsilon$$

Enables jumping timesteps for fast 10–50 step sampling.

2. The Coordinate Scaling Fix



Why Rescaling is Essential: Without dividing by $\sqrt{\bar{\alpha}_t}$, mode centers shrink over time, making cross-timestep KDE comparisons invalid.

Work Done Before Prep Review: 2D Toy Oracle & Codebase

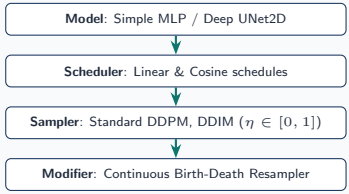
8-Mode Gaussian Ring Oracle

- Synthetic 2D ring ($R = 2.0$, $\sigma_0 = 0.15$):

$$p_0(x) = \frac{1}{8} \sum_{m=1}^{8} \mathcal{N}(x; \mu_m, \sigma_0^2 I)$$

- **Analytical Truth:** Hand-derived exact formulas for noisy density $p_s(z)$ and score $\nabla \log p_s(z)$.
- **Zero Model Error:** Provides an exact playground to isolate algorithm mechanics without neural net bugs.

Modular PyTorch Architecture



Initial Finding: Confirmed Birth-Death accelerates stationary Langevin dynamics on multimodal targets before review.

Work Done After Prep Review: The Log-Density Roadblock

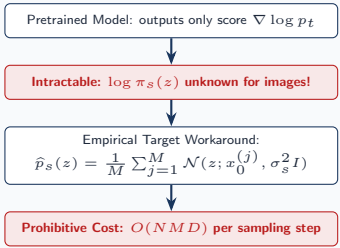
The Structural Roadblock on Real Images

- **Birth-Death Requires:** Log-density evaluation:

$$\beta_i = \log \widehat{q}(z_i) - \log \pi_s(z_i)$$

- **Diffusion Models Provide:** Only the score vector $\nabla \log p_t(x_t)$ via noise prediction.
- **Intractable Normalizer:** The partition function $Z_t = \int \tilde{p}_t(x)dx$ is completely unknown in high dimensions.

Flowchart of the Real-Data Bottleneck



Work Done After Prep Review: Standardized Metrics & Sweeps

Deliverable 4 Evaluation Metrics

- **FID (Fréchet Inception Distance):** Image realism compared against 500 validation images (lower is better).
- **Unique Ratio (U):** Diagnoses mode and particle collapse:

$$U = \frac{\text{Unique Generated Images}}{\text{Total Images Generated}} \in (0, 1]$$

- **MMD^2 & NLL:** Accurate 2D distribution matching.
- **Wall-Clock Time:** CUDA-synchronized sampling runtime.

Systematic Sweep Parameters

- **Sampling Steps (S):** $S \in \{5, 10, 25, 50\}$.
- **BD Budget (τ):** $\tau \in \{0.01, 0.1, 1.0\}$.
- **Intervention Frequency (k):** Every $1, 2$, or 5 steps.
- **Stochasticity (η):** $\eta = 0.0$ (deterministic DDIM) vs. $\eta = 1.0$ (stochastic Langevin).
- **Batch Size:** $N = 16$ with fixed reference set M.

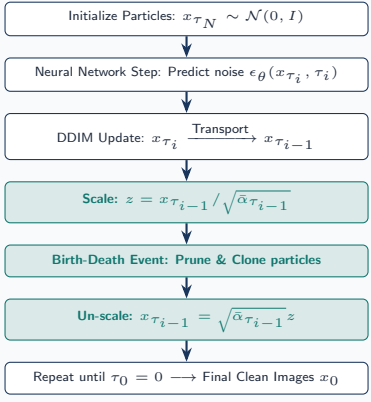
Proposed Approach: The Operator-Splitting Sampling Loop

Two Decoupled Operations

- **1. Temporal Transport (DDIM):** Advances particles across timesteps $\tau_i \rightarrow \tau_{i-1}$, peeling away noise along the time axis.
- **2. Spatial Resampling (BD):** Shuffles particles at a fixed noise level τ_{i-1}, duplicating promising modes and killing low-density ones.

Key Realization: DDIM moves particles *in time*; Birth-Death moves particles *in space*.

Complete Sampling Flowchart



Proposed Approach: The Birth-Death Branching Mechanism

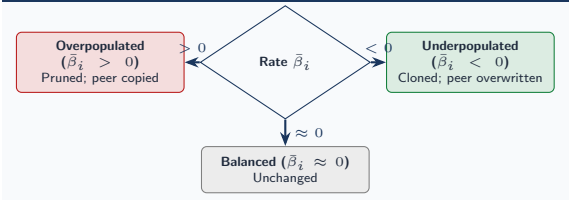
Discrepancy Rate Formulation

- **Particle Density (KDE):**

$$\log \widehat{q}_h(z_i) = \mathrm{LSE}_{j=1}^{N}\left(-\frac{\|z_i - z_j\|_2^2}{2h^2}\right) - \log N$$

- **Discrepancy Rate:** $\beta_i = \log \widehat{q}_h(z_i) - \log \pi_s(z_i)$
- **Mean-Centering:** $\bar{\beta}_i = \beta_i - \frac{1}{N}\sum_j \beta_j$ keeps total particle count N fixed.
- **Event Probability:** $p_i = 1 - e^{-|\bar{\beta}_i|\Delta\tau}$.

Branching Decision Flowchart



Vectorized PyTorch GPU scatter/gather index operations execute in < 1 ms.

Proposed Approach: Neural Network Architecture

Diffusers UNet2DModel Backbone

- **Hourglass Feature Hierarchy:** Downsampling blocks with residual layers $\rightarrow$ bottleneck $\rightarrow$ upsampling blocks with skip connections.
- **Capacity Settings:**
  - CIFAR-10 (32×32): $[128, 256, 256, 256]$ channels.
  - CelebA-HQ (256×256): $[128, 128, 256, 256, 512, 512]$ channels.
- **Spatial Self-Attention:** Embedded at 16×16 resolution.
- **Timestep Conditioning:** Sinusoidal embeddings injected into every residual block.

Training Objective

- Standard Mean Squared Error (MSE) on noise:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]$$

- **Key Implication:** The model learns only score gradients $\nabla \log p_t$, never the partition function or density magnitudes.

Datasets and Preprocessing

1. 2D Toy Ring ($D = 2$)

- Synthetic 8-mode Gaussian mixture.
- Exact closed-form scores and densities.
- **Purpose:** Direct visual scatter validation of particle trajectories.

2. CIFAR-10 ($D = 3,072$)

- 60,000 images ($32 \times 32 \times 3$).
- 10 object classes; $[-1, 1]$ scaled.
- Reference batch $M = 256$.
- 500 test images for standardized FID computation.

3. CelebA-HQ ($D = 196,608$)

- 30,000 celebrity face images.
- High-res ($256 \times 256 \times 3$); $[-1, 1]$.
- Reference batch $M = 128$.
- Standardized 500-sample FID benchmark on cloud GPUs.

Dimensionality Spectrum: Spans 5 orders of magnitude ($D = 2 \longrightarrow D = 3,072 \longrightarrow D = 196,608$).

Experimental Setup and Details

Compute Hardware and Environment

- **Cloud Accelerator:** NVIDIA Tesla T4 GPU (16 GB GDDR6).
- **Host System:** Intel Xeon @ 2.20 GHz (4 vCPUs), 32 GB RAM.
- **OS & CUDA:** Ubuntu 22.04 LTS, CUDA 12.1.
- **Software Stack:** PyTorch 2.4, Hugging Face `diffusers`, `torchmetrics` Inception-v3 pool3.

Execution Protocol

- **Reproducibility:** Fixed seed ($seed = 42$) across baseline and birth-death runs.
- **Local Datasets:** Datasets downloaded locally into `data/` before evaluation to eliminate network jitter in timing.
- **Timing Accuracy:** Measured using synchronized CUDA event timers.

Results I: Synthetic Validation – Does Birth-Death Work?

1. Langevin Mixing: Confirmed ($5\times$–$10\times$ speedup)

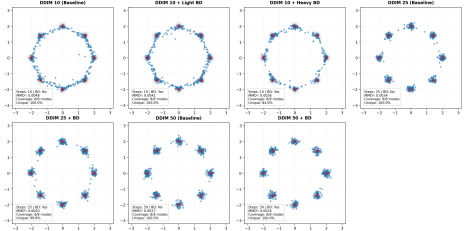
N	Method	Steps	Total Var (↓)	Speedup
64	Baseline / **BD**	5k / **5k**	0.614 / **0.345**	$2.0\times$
128	Baseline / **BD**	26k / **5k**	0.223 / **0.223**	$\mathbf{5.2}\times$
1000	Baseline / **BD**	20k / **5k**	0.313 / **0.089**	$\mathbf{10}\times$

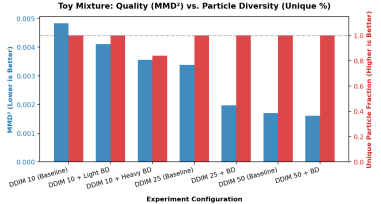
- BD is a genuine mixing accelerator for stationary Langevin dynamics with known target densities.

2. But in Diffusion: More Steps Always Win

Steps	η	Config	MMD^2	Uniq	Time
5	0	Baseline	0.0408	1.00	5 ms
5	0	BD (0.7, 0.1)	0.0378	0.84	10 ms
10	0	Baseline	**0.0060**	1.00	**4 ms**
10	0	BD (0.3, 0.1)	0.0057	0.93	18 ms

- 10-step baseline (4 ms) beats 5-step BD (10 ms) by $\mathbf{6}\times$ and runs $2.5\times$ faster!





Results II: High-Dimensional Diagnostics & Biased Score Repair

Diagnostic: High-Dim Distance Concentration

Null Test: 1,024 exact Gaussian samples, 1 BD step:

$\mathbf{d}$	Self-Weight	Unique Ratio	MMD Error
2	8.5%	0.82	$0.0003 \rightarrow 0.0026$
8	53.1%	0.61	$0.0003 \rightarrow 0.0042$
32	**100.0%**	0.51	$0.0003 \rightarrow 0.0024$
768	**100.0%**	**0.45**	$0.0003 \rightarrow$ **0.0012**

- In high dimensions, all pairwise distances concentrate.
- Every particle sees only itself (100% self-weight), blinding the density estimator into arbitrary duplication.

The Positive Finding: Fixing Biased Scores!

Deliberately corrupted score: $s_{\text{model}} = c \cdot s_{\text{true}}$ (10 steps):

Multiplier	Base MMD	BD MMD	Base NLL	BD NLL
$c = 1.0$ (Normal)	0.0062	0.0052	2.29	1.20
$c = 0.5$ (Severe Bias)	0.1667	**0.0260**	235.8	**6.47**
$c = 0.7$	0.0131	0.0064	5.37	2.57

- Under severe model bias ($c = 0.5$), BD reduces distribution error by $\mathbf{6.4}\times$ and NLL by $\mathbf{36}\times$!
- **Conclusion:** Birth-Death is an **error corrector**, not an inference step reducer.

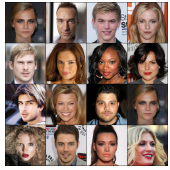
Results III: CIFAR-10 and CelebA-HQ Benchmarks

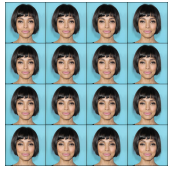
Quantitative Benchmarks (NVIDIA Tesla T4)

Dataset & Setting	Steps	Base FID (↓)	BD FID (↓)	Unique (U)
CIFAR-10 (32×32)	10	**144.74**	150.21	1.0000
CIFAR-10	25	**135.73**	136.55	1.0000
CIFAR-10 (Heavy BD $\tau = 1$)	25	**135.73**	224.83	**0.5625**
CelebA-HQ (256×256)	10	**65.64**	81.63	1.0000
CelebA-HQ	25	**66.02**	87.47	0.8125
CelebA-HQ ($\eta = 0.0$)	25	**66.02**	98.33	**0.0625**

Catastrophic Duplicate Locking in DDIM ($\eta = 0$)

In deterministic DDIM, transitions are bijective. Once duplicated, particles share identical coordinates forever ($U = 0.0625$, 15 of 16 faces collapsed into clones).





CelebA Faces: Baseline ($U = 1.0$) vs. Deterministic BD ($U = 0.06$)





CIFAR-10: Baseline ($U = 1.0$) vs. Heavy BD ($\tau = 1.0$)

Conclusions: What We Learned from This Project

1. Where Birth-Death Works

- **Stationary Langevin Mixing:** Confirmed $5 \times -10\times$ acceleration when target density is tractable.
- **Biased Score Repair:** Cuts distribution error by $6.4\times$ when network gradients are systematically biased.

2. Why It Fails to Accelerate Diffusion

- **No Log-Density in Models:** Requires repeated intractable log-density evaluation at every step.
- **High-Dim Collapse:** Distance concentration destroys KDE accuracy.

Fundamental Distinction: Transport vs. Resampling

Temporal Transport (Diffusion Steps):
Moves particles *across noise levels* in time ($x_t \rightarrow x_{t-1}$).
Essential to reach clean data.

↓

Spatial Resampling (Birth-Death):
Shuffles particles *at fixed noise level* in space.
Cannot advance particles across timesteps!

Takeaway: If particles land in a bad region, BD can only clone bad particles; it cannot transport mass forward in time.

Plan for Novelty Assessment: Concrete Future Extensions

1. Replace KDE with Transition Kernels

- **Problem Solved:** High-dimensional KDE distance concentration (100% self-weight).
- **Proposed Method:** In Sequential Monte Carlo (SMC), weight particles using the model's analytical reverse transition density:

$$p_\theta(x_{t-1}|x_t)$$

- **Advantage:** Completely eliminates empirical Euclidean distance matrices and self-domination!

2. Exact Density Normalizing Flows

- **Problem Solved:** Missing marginal log-density $\log \pi_t$ in diffusion models.
- **Proposed Method:** Evaluate Birth-Death on models with exact tractable log-densities:
  - Continuous Normalizing Flows (CNFs)
  - Glow architectures
- **Advantage:** Rigorously decouples score error from density estimation error.

Roadmap of Proposed Extensions



References: Foundational Literature

1. Jonathan Ho, Ajay Jain, and Pieter Abbeel. *Denoising Diffusion Probabilistic Models*, Advances in Neural Information Processing Systems (NeurIPS), 2020. https://arxiv.org/abs/2006.11239

2. Jiaming Song, Chenlin Meng, and Stefano Ermon. *Denoising Diffusion Implicit Models*, International Conference on Learning Representations (ICLR), 2021. https://arxiv.org/abs/2010.02502

3. Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. *Score-Based Generative Modeling through Stochastic Differential Equations*, International Conference on Learning Representations (ICLR), 2021. https://arxiv.org/abs/2011.13456

4. Jianfeng Lu, Yulong Lu, and James Nolen. *Accelerating Langevin Sampling with Birth-Death*, Annals of Applied Probability, 2019. https://arxiv.org/abs/1905.09863

5. Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. *DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps*, Advances in Neural Information Processing Systems (NeurIPS), 2022. https://arxiv.org/abs/2206.00927

All references strictly reflect the foundational papers cited and evaluated in the project report.